

REPORT SUGGESTIONS O ADVICES BRISTILE 2023

(300123 to 030223) DANDENONG

SANDS

***In my evaluation, the optimum sand for the production should be of the following granulometry.**

n° Sieve.	Size in mm.	%	% of the total
4	4,76	0	0
8	2,38	10	10
12	1,7	17	7
116	1,19	27	10
30	0,59	57	30
50	0,297	78	21
100	0,149	94	16
200	0,074	97	3
Powder	0,05		3
			100

***PREFIXED OBJECTIVES :**

Fineness Modul (2,85).....(2,7 a 3)

Fines 1 (5%).....(2 a 5 %)

Sand Equivalent (2,85).....(80 al 85 %)

SAND WEIGHING HOPPER

***Sand weighing Hopper,to consider discharges on a discontinuous basis to mixer, to reduce dirt in the mixer.**

***Eliminate crushing sand screws, which are currently not functioning.**

CEMENT WEIGHING HOPPER

***Cement weighing Hopper (Eliminate pneumatic vibrator hopper by an impact vibrator).**



***The sand/cement ratio, that I have used it in factories is**

"3,65 to 4,09"

ADDITIVES

WATERPROOFING :

***To reduce or avoid impermeability problems in concrete tiles, try addingadd in the mixture "Calcium Stearate".**

PLASTICIZING

***DO NOT use chemical products, powders or liquids in the tile mix that can alter the structural characteristics (to achieve a greater initial resistance or a greater plasticity of the mortar) (accelerators, stiffeners, setting retarders, etc.) so far in my experience I have not identified that provide more advantages, on the contrary if I have identified disadvantages in its use. There may be other criteria in this regard, which I respect ,”I do not have experience”.**

***In factories for surface finishes such as slurry YES,(“summer < +25º”) small doses are used for the formation of a 1 mm layer and alter its characteristics.**

WATER MIXER.

***Hot water for Winter by (heat exchanger) or alternative water heating system to the mixer, to achieve a minimum temperature of mixture 20 to 25 ºC.**



***Depending on the properties of the water in the settling tanks, the water can be reused in the mixture process.**



MIXER SITUATION.

***Excessive dirt in all areas of the mixer, too much material left on top/botton of the mixer satellites.**



CLEANER

The cleaning mode is manual ,a lot of effort and time (I've only seen dry cleaning?)with pneumatic hammer and other tools.



***I consider that the cleaning should be complemented by cleaning with the high pressure pump and the application of the liquid, thereby avoiding the incrustations. Zetolan MEK 3.**

ZETOLAN MEK 3



LeycoChem
NORSEKEM
Product groups)
LeycoChem Leyde Plant Protection Plant
Cleaning

SKU/CODE: MEK3

PROTECTION FOR CONCRETE MIXERS AND CONSTRUCTION EQUIPMENT
ZETOLAN MEK 3 forms an effective adhesive protective film on the surfaces on which it has been sprayed, preventing the adhesion of residual deposits of concrete and mortar. Deposits can be removed by means of a water jet. As soon as the protective film wears off, a new protective layer of ZETOLAN-MEK 3 can be applied to the wet surface after cleaning.

Advantages:

- Costly contamination of essential parts and grease grooves is avoided.
- Longer service life of the machinery due to better care.
- Reduces labour cost for cleaning.
- Does not damage rubber, glass, metal, paint, wood or clothes.
- Economical application, since only a thin film is needed for protection.

Imágenes de zetolan mek



“Once the inside of the mixer has been cleaned, it is advisable to impregnate this liquid with the exterior of all the elements that may come into contact with the concrete of the mixer, thereby avoiding the incrustations and the ones that are generated can be easily cleaned.”

***In order to channel the water generated during the cleaning of the mixer, it would be necessary to make a channel in the floor from the concrete mixer to the settling basins.**

HOW TO MAKE THE MIXER LESS DIRTY.

***A complementary objective is how to reduce the creation of dirt, in the operation of adding sand, cement and pigment.**

***The sand should be discharged pulsatingly into the mixer, once the sand discharge is finished, the cement should be discharged, I recommend to extend the piping inside the mixer with a rubber sleeve.**

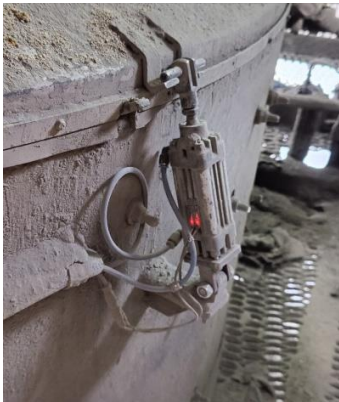
***Is necessary water supply (some nozzles) to the inside of the mixer, in order to” clean the mixer” as much as posible on the feed so that they are above the satellites mixer.**





MIXER SAFETY

***Although there are safety devices to prevent the mixer from starting instantly, I consider that it would be necessary to install 3 more devices, to prevent it from not being possible to**



***Fit safety locks on the 3 cleaning access covers.**



MOISTURE MIXTURE

***Install MOSITURE CONTROL system at the base of the mixer, I recommend HIDRONIX VI or similar.**

HOPPER MIXER

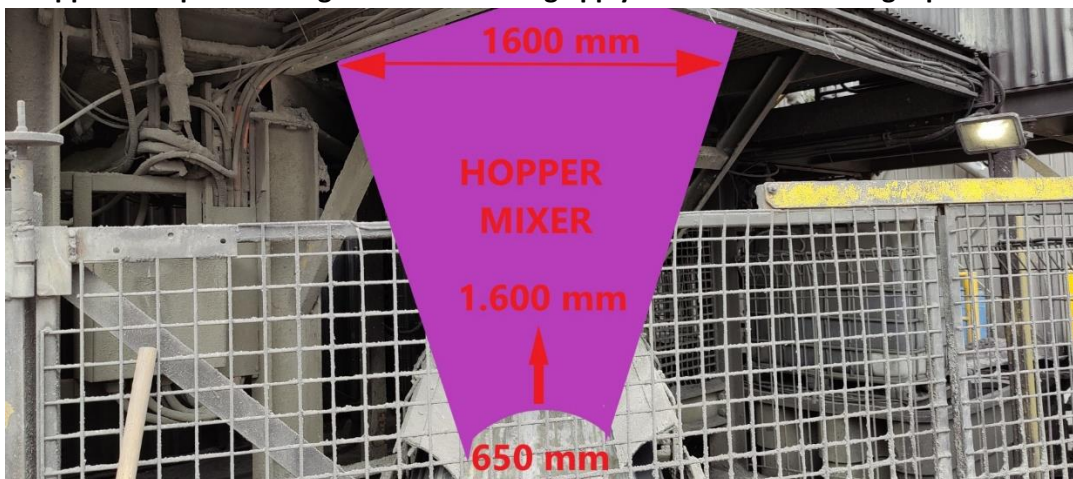
Excessive dirt, a lot of effort and time

*It should not be round, it should be straight like the general factory hopper.



*Consider installing a longer straight hopper (measurements are approximate) avoid impacts and retention of concrete in the hopper.

*Hopper with plastic lining and after cleaning apply “Zetolan” antifouling liquid.



DUST COLLECTION FILTER CONCRETE MIXER

***A dust collector would have to be installed in the concrete mixer and at other points where necessary.**



MIXER AREA

***The entire working area of the mixer, floors and side enclosures would have to be reconditioned.**



BELT CONVEYOR MIXER TO CENTRAL HOPPER.

***Possible accommodation of the scraper at the front, in the middle area of the lower part of the traction roller with tensioners. (If it works well, there is no need to change)**



Overhaul of the belt base rollers.



Anti-fall system of the 2 lower belt rollers.



Cover belt lower part of the belt in the passage of persons.(Safety)

HOPPER CENTRAL FACTORY

***Very closed at the bottom, I think that should be wider with more vertical sides due to the width of the belt.**



BELT HOPPER CENTRAL TO EXTRUDER.

The position of the front roller is set back to facilitate the removal of the hopper and extruder head.



***I think that the belt is possible deflected to the right, possible inspection by the maintenance mechanic.**